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Exchange market pressure and primary commodity – exporting emerging markets

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ABSTRACT

This paper studies short-term sensitivity between exchange market pressure and various domestic and external factors in primary commodity-exporting emerging markets. The paper focuses on the top country-commodity groups in sugar, cereal, fuels, ores and coffee during the pre-peak and post-peak commodity price periods across floating and pegged exchange rate regimes, using the price of crude oil as a general benchmark. Employing a panel model and panel VAR analysis, the paper finds the heterogeneity of response patterns unique to country-commodity groups and exchange rate regimes. According to the results, in flexible regimes, volatility occurs via the foreign exchange market, interest rates, and domestic credit cycles, feeding into the social costs for structurally weaker economies. Hard exchange-rate pegs often result in a drain on international reserves as the terms of trade deteriorate following post-price peaks, leading to unpopular depreciation. These results accentuate concerns over uneven international trade patterns, an open economy's short-term foreign exchange policy, and speculative capital flows. Such sensitivity has broad implications for macroeconomic balance and the sustainability of implied exchange rate targets in the presence of a foreign exchange constraint across emerging markets.

KEYWORDS

Commodity dependence; emerging markets; exchange market pressure; foreign exchange constraint

JEL CLASSIFICATION

E58; F31; F40; F41

I. Introduction

This paper studies short-term sensitivity in the foreign exchange regimes of the primary commodity-exporting emerging markets, classified as commodity-dependent countries (CDC) by UNCTAD (2014), before and after commodity price peak periods. Those CDCs benefited from windfall revenues after the 2007 global financial crisis (GFC) through their high-priced primary commodity exports. Many accumulated foreign currency-denominated debt, leveraging the transient ability to borrow at historically low-interest rates (e.g., Guscina, Pedras, and Presciuttini 2014). Yet, with the decline of global commodity prices in mid-2014, the evolving patterns of international trade, and the monetary tightening in advanced economies, emerging markets are now facing liquidity pressures shadowed by a familiar foreign exchange constraint. A combination of domestic and external factors exerts pressures on national currencies, impacting the trade off between primary commodity exports and manufactured

imports. This has implications for macroeconomic balance in the structurally weaker economies.

In such a context, this paper uses exchange market pressure (EMP), the sum of domestic currency depreciation and the international reserves loss (e.g., Aizenman and Hutchison 2012; Girton and Roper 1977), to capture impacts from macroeconomic and financial factors in relation to currency pressures under varying exchange rate regimes. Offering an informed, real-time, insight the EMP is commonly applied in the emerging markets literature (e.g. Aizenman, Lee, and Sushko 2012; Frankel 2009). A positive change in EMP – likely due to a currency depreciation or a loss of international reserves – reflects a net excess demand for foreign currency. Therefore, EMP is especially helpful for this paper's attempt at exploring the extent to which CDCs have adopted implicit exchange rate targets within domestic macroeconomic balance.

This paper is motivated by and contributes to the earlier studies on exchange market pressure (e.g., Aizenman, Chinn, and Ito 2015; Frankel and

Saravelos 2010; Rose and Spiegel 2009) that have endeavoured to explain countries' vulnerability in relation to key macroeconomic and financial indicators. There is an evident diversity of approaches (e.g., Aizenman and Hutchison 2012; Frankel 2009; Tanner 2002, 2000) with variations of short-term indicators across the analyses. More recently, Aizenman and Binici (2015) added the problem of capital controls and policy signals from the US Federal Reserve to their analysis of exchange market pressures across selected Organization for Economic Co-operation and Development (OECD) and emerging market groups. With an exhaustive literature review on EMP, Fiador and Biekpe (2015) also make an explicit connection between EMP, monetary policy and external trade factors in the economies of Sub-Saharan Africa.

The paper makes three specific contributions to the broad emerging markets literature. First, this paper analyses the EMP and its relevance as an analytical tool for macroeconomic policy guidance across the CDCs within specific commodity groups. This is important due to a diverse compilation of each group and of the leading export commodity's varying effects on the respective economies' domestic and external balances. A panel study approach appears to be common across the literature.

Second, the paper relies on a panel VAR estimation to supplement panel results (following Love and Zicchino 2006; and Abrigo and Love 2015). In it, the application of panel VAR helps explore deeper interdependencies across variables that are not immediately obvious, including decomposing the EMP between exchange rate and international reserves changes. Such decomposition allows the capture of minute commodity group effects in response to shocks to the main explanatory variables. In addition, panel VAR allows the incorporation of time-varying coefficients and specific shocks analysis, with dynamic cross-sectional heterogeneities (see Canova and Ciccarelli 2013 for informed overview of panel VAR models).

Third, the paper documents the differences in country-commodity groups' EMP responses to short-term domestic and external stresses across adopted (or implicit) exchange rate regimes within

each group. Dynamics associated with dominant primary commodity export also drives a broad range of responses in dealing with foreign exchange constraint and volatility, among others.

Following this introduction, Section 2 summarizes some relevant facts about commodity dependent emerging markets and develops a core analysis sample group. Section 3 presents the paper's methodology and data. Section 4 goes over results of the panel study approach, followed by a relatively new Stata routine – pvar via panel vector autoregression (VAR) model estimation. Section 5 concludes by reviewing the main results of foreign exchange management across different commodity groups and macroeconomic development implications.

II. Emerging markets: stylized facts across commodity-exporting groups

It is useful to first review some key characteristics of post-GFC emerging markets. Figure 1 tells the by now known story of the commodity super-cycle. The trend, measured by the 'All' index, reflects an increase until mid-2011, stabilization through mid-2013 to early 2014, and from then a subsequent decline in the price index. The energy index (i.e. mainly the oil prices) may serve as a guiding benchmark for correlation across other commodities.

The higher correlation between primary commodities' prices may be driven by technological evolution, including the search for a renewable source of energy. There may be stronger price shock transmissions as substitutability between non-renewables and renewables rises (e.g., Gevorkyan 2017b; Canuto 2014; Dobbs et al. 2013). At the same time, the proliferation of commodity derivatives on financial markets has contributed to more recent volatility in commodity prices (Arezki et al. 2013). Finally, despite the overwhelmingly common trend, Figure 1 also reveals a difference in post-GFC peak prices for different commodity groups, as we explore below.

Helping to narrow down the country focus of this paper, Table 1 advances five explanatory points. First, the sample is constructed around five major primary commodity groups. Here, the fuels group includes crude oil, natural gas and

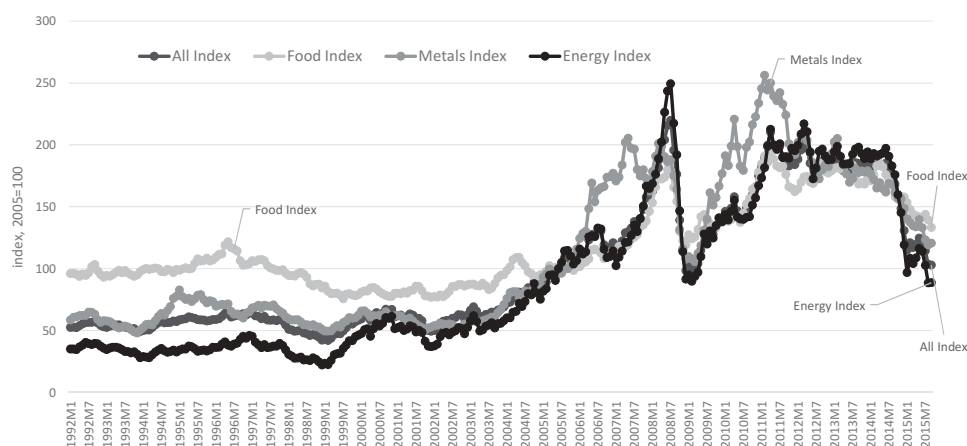


Figure 1. Primary commodities price indexes, 2005 = 100.

Source: IMF (2016).

Note: All Commodity Price Index, 2005 = 100, includes both fuel and non-fuel price indices.

other minerals. The ores group, or metals, comprises iron ore, copper, nickel and other metal exporters and prices. The five commodity groups offer a general enough framework capturing key CDC's exports. For brevity, this paper's econometric analysis focuses only on sugar, ores and fuel exporters.

The second explanatory point concerns the country choice in each commodity group. The data on an individual country's shares in global commodity exports for the 1995–2014 period in each respective group are derived from UNCTAD's online statistical database (UNCTAD 2016). Excluding advanced and OECD member economies (as well as China in ores, and the Persian Gulf states in fuels) allows us to sort for emerging and developing countries, focusing on the respective commodity group. Only the top five exporters per group, sorted in descending order based on their specific commodity exports as share of the global market, appear in Table 1. Each of the selected top five exporters is classified as a commodity-dependent country according to the UNCTAD (2014).

The third observation is a visible decline in the growth rate of gross domestic product (GDP) per capita in the post-GFC period (column 4 in Table 1) compared to the 14-year period for which the World Bank data are available. On average for the entire sample, the annual growth rate for GDP per capita was 2.8 per cent in the period 2010–2014 compared to 3.13 per cent in 2000–2014. We

consider 2010 to be the first post-GFC year in the emerging markets sample because, according to data, it is the first year to register positive growth. Among the notable exceptions are strong positive growth rates in Côte d'Ivoire and Ghana, which seemed to have benefited from foreign capital inflows at the post-GFC peak of commodity prices.

The fourth note on Table 1 is about the positive change in international reserves. For the entire sample group, international reserves averaged 21.4 per cent of gross national income (GNI) for the period 2010–2014. The major CDCs have benefited from high commodity prices, feeding into domestic currency appreciation and a build-up in international reserves. Accumulated reserves, channelled in sovereign wealth (reserve) funds, may play the role of a stabiliser in a sudden reversal of capital flows, supporting a competitive (or policy-targeted) exchange rate. Related, Dominguez, Ito, and Hashimoto (2012) find that higher accumulation of international reserves also contributes to a more robust, post-crisis recovery.

Finally, Table 1 also shows the diversity in exchange-rate regimes. For example, between the sugar and ores exporters group, the dominant regime is floating. Here, the freely floating one is the most liberal regime, while a crawling peg is classified as a soft peg (IMF 2015). The picture is different in the fuels group, in which soft pegs seem to dominate the group. Aside from some country overlaps (e.g. Brazil and Russia),

Table 1. Select primary commodities exporters in emerging markets.

	Country	Share of global exports, %	Post GFC GDP per capita loss	Post GFC IR change	FX Regime	Cereal	Country	Share of global exports, %	Post GFC GDP per capita loss	Post GFC IR change	FX Regime
Sugar	Brazil	16.3%	0.15	0.04	Floating	Cereal	Argentina	4.6%	0.91	(0.03)	Crawl-like
	Thailand	5.5%	(0.04)	0.08	Floating		Thailand	3.8%	(0.04)	0.08	Floating
	India	1.8%	0.43	(0.005)	Floating		India	2.9%	0.43	(0.005)	Floating
	Colombia	1.7%	0.65	0.002	Floating		Russia	1.9%	(2.29)	0.01	Floating*
	Guatemala	1.5%	0.27	0.004	Crawl-like		Ukraine	1.8%	(2.35)	(0.01)	Floating*
Ores	Chile	5.5%	0.65	(0.01)	Free floating	Coffee	Brazil	7.3%	0.15	0.04	Floating
	Russia	4.4%	(2.29)	0.01	Floating*		Côte d'Ivoire	5.0%	2.69	0.03	Conventional
											peg
Fuels	Brazil	3.7%	0.15	0.04	Floating		Colombia	3.6%	0.65	0.002	Floating
	South Africa	3.1%	(0.75)	0.03	Floating		Ghana	2.8%	2.12	0.02	Floating
	Peru	1.7%	0.46	0.07	Floating		India	2.8%	0.43	(0.005)	Floating
	Russia	9.7%	(2.29)	0.01	Floating*						
	Nigeria	3.1%	(2.14)	(0.10)	Other						
Venezuela					managed						
			(1.68)	(0.07)	Conventional						
					peg						
Algeria			(0.73)	0.22	Other						
					managed						
Angola		1.5%	(4.37)	0.09	Crawl-like*						

Source: UNCTAD Stat (2016) for commodity shares, World Development Indicators (WDI) (2016) for GDP and IR estimates, UNCTAD (2014) for CDC designation, IMF (2015) for exchange rate regimes, and author's estimates.

Note: Fuels group excludes the Persian Gulf states; ores group does not include China; Chile is accounted for despite its recent OECD membership. GDP per capita growth is annual percentage change from WDI database. IR = international reserves as a share of gross national income (GNI). FX regime = exchange rate regime in 2015. * 2014 switch from another regime. GFC = global financial crisis, GDP = gross domestic product; IR = international reserves; FX regime = exchange rate regime.

subsequent analysis shows how the fuels group's monetary authorities are maintaining an implicit fixed exchange rate target, which helps to explain the differences across commodity groups.

Adding to the complexity of commodity-dependent exports is high accumulation of foreign currency denominated debt and volatility in capital flows. For example, in their analysis of sub-Saharan African exporters (SSA), Gevorkyan and Kvangraven (2016) show that instability in commodity-pegged revenue detrimentally affects the ability to tap into international capital markets at competitive borrowing rates (also Dafe, Essers, and Volz 2017). Senga, Cassimon, and Essers (2018) extend the analysis to the country-specific variables, further prompting this paper's choice for country-commodity grouping. Separately, Hegerty (2013) connects these concerns with exchange market pressure analysis for West African commodity exporters, finding significant regional spillovers.

The effect of declining capital inflows (Figure 2), since commodity prices decline, is visible in sharp reversals of the foreign direct investment (FDI). On average, in the post-GFC period, external debt as a share of GDP, with a few exceptions, has also declined. This is more of a reflection of nominal GDP decline, while short-term debt is still being added to the balance sheet, as is implied in some

analysis (e.g. Canuto 2015; IIF 2016; WDI 2016). Unless there is some recovery in capital inflows, a continuous loss of foreign exchange may be leading to a disproportionate drain on international reserves under exchange rate pegs and in more direct structural impacts in CDCs with flexible regimes. In both cases, sovereign debt sustainability becomes of concern, feeding into further self-fulfilling negative expectations in the market (e.g. Aizenman, Lee, and Sushko 2012 on the short-term implications of reserves loss, commodity prices volatility and exchange market pressures).

III. Empirical methodology and data description

Methodology

As a baseline approach to analysis of exchange market pressures across different commodity exporters, we construct the following panel model:

$$EMP_{it} = \alpha + \beta_i DM_{it} + \theta_i EX_{it} + \varepsilon_{it} \quad (1)$$

where EMP is the exchange market pressure index for commodity group i at time t , DM is a group of domestic factors, and EX is a group of external explanatory variables (as specified in Table 2). The error term is i.i.d. and represented by ε_{it} .

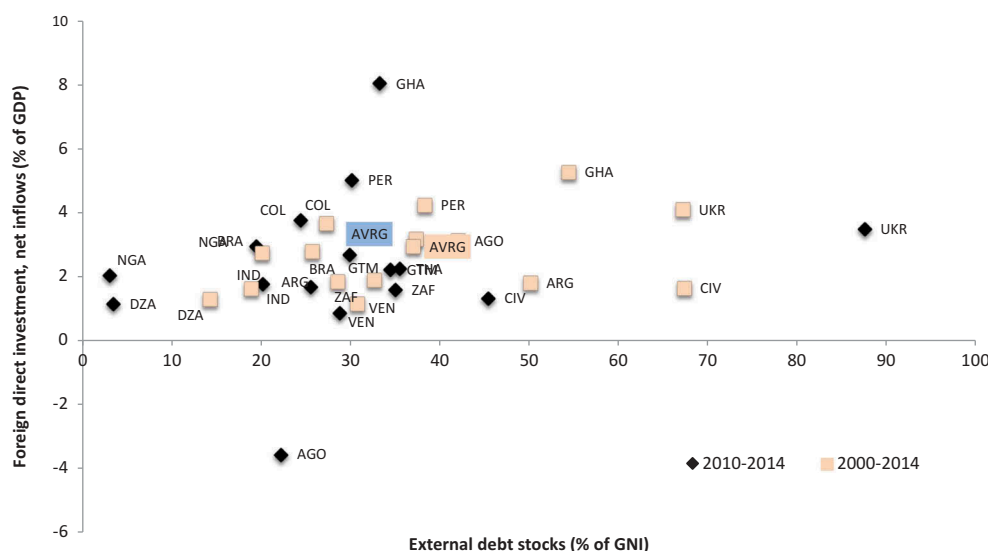


Figure 2. Changes in FDI and external debt in emerging markets, 2000–2014.

Source: World Development Indicators (2016) and author's calculations.

Note: See country code key in Appendix Table A1.

Table 2. Variables, expected signs, and data sources.

Variable	Description	Source	Expected sign
Dependent variable			
EMP	Difference between % depreciation in local currency per USD rate and % international reserves loss	WB GEM	
Explanatory variables			
<i>Domestic</i>			
MPC	Industrial Production, USD seasonally adjusted	WB GEM	(-)
BIDASK	Bid-Ask spread [local currency per USD]	Oanda.com	(+)
LR	Domestic lending rate	IMF IFS	(+)
<i>External</i>			
T10	US T10-year bond yield, const. maturity	FRED St. Louis	(+)
VIX	Volatility index	FRED St. Louis	(+)
OIL	Per cent change in average crude oil price	UNCTAD Commodity Prices	(-)
GOLD	Per cent change in average gold price	UNCTAD Commodity Prices	(-)
SUGAR	Per cent change in average sugar price	UNCTAD Commodity Prices	(-)
ORES	Per cent change in average ores price	UNCTAD Commodity Prices	(-)

For now, we follow the literature (e.g., Aizenman and Hutchison 2012), and our standard EMP definition in general form appears as:

$$EMP_{it} = \frac{e_{it} - e_{it-1}}{e_{it-1}} - \frac{R_{it} - R_{it-1}}{R_{it-1}} \quad (2)$$

where e is the exchange rate (domestic currency per US dollar), and R is international reserves (less gold). Equation (2) defines EMP as a measure of a monthly relative change in exchange rate and international reserves.

Some extensions to the standard definition include relative interest rate differentials (as proxy for arbitrage flows), deflating international reserves by monetary base, or developing a standardized difference deflating by the standard deviation of respective variables (e.g., Aizenman and Binici 2015; Klaassen and Jager 2011; Tanner 2002). In an EMP analysis of a small open transition economy, Gevorkyan (2017) derives a cross-EMP gauge for a case of dual currency targets (US dollar and Russian rouble, in the case of Armenia). Baig, Narasimhan, and Ramachandran (2003) carry out their EMP analysis for India, revealing the monetary authority's preference for currency depreciation and conservative net selling of international reserves in crises only.

Gochoco-Bautista and Bautista (2005) employ a VAR model to explore the role of domestic credit growth and the interest-rate differential in relation to

the EMP. The authors find that having a contractionary policy reduced EMP in non-crisis periods. However, the evidence of the 1997 Asian crisis showed that during the crisis higher interest rates led to a higher EMP. For a more complete literature review on EMP and range of empirical approaches, see Aizenman and Binici (2015) and Tanner (2002).

Consistent with the cited literature and in line with earlier work by Girton and Roper (1977), we follow the baseline EMP definition provided in equation (2). Without a loss of generality, as found in Aizenman and Binici (2015) or Hegerty (2013), a variety of EMP measures, nuanced by specific factors, tell a story compatible with baseline EMP. The literature on exchange rates determination (e.g. earlier cited work; Chen and Chou 2015; and Coudert et al, 2015; on commodity currencies for literature review therein) guides the choice of some short-frequency domestic factors (DM) and external (EX) factors. It is also important to keep in mind data limitations associated with specific frequency and country sample.

Therefore, in this study, the DM matrix includes the monthly domestic industrial index (MPC), domestic lending rate (LR), and the interbank bid-ask spread for the US dollar (USD) expressed in local currency (BIDASK). Intuitively, it is reasonable to expect a negative correlation between the MPC (a possible proxy for domestic economic activity) and the EMP. As industrial output structurally improves,

so does the economy's currency (an appreciation is a reduction in the exchange rate change component of the EMP), with possibly a larger proportionate gain in international reserves pushing the overall EMP down.

Similarly, heightened risk perception or foreign exchange market volatility due to either shaky macro-economic foundations or a function of the country's financial deepening may be pushing the EMP index higher. Hence, a positive change in either BIDASK or LR might be a sign of macroeconomic stress (the LR in addition to the banking sector's reaction also capturing inflationary pressures) leading towards reserves loss or currency depreciation in structurally smaller economies. The challenge is to parse the commodity group dynamic and address country-specific tendencies. The intuition for both BIDASK and LR is consistent with the findings of Gochoco-Bautista and Bautista (2005). Also, Huang and Masulis (1999) posit that the bid-ask spread may be a suitable proxy for competitive pressures in the foreign exchange spot markets, illustrating the concern over self-feeding tendencies in highly volatile non-linear markets of the CDCs.

The external factors (EX) group consists of the 10-year US Treasury bond with constant maturity (T10), volatility index (VIX), and individual commodity spot process (OIL, SUGAR, ORES). Both T10 and VIX offer a sensible gauge of external market volatility. In particular, the VIX may carry impulses from the speculative capital flows entering or leaving emerging markets, in particular, first-time borrowers in international capital markets. As such, the VIX serves a better proxy for the global investors' preferences and perceptions. Hence, the expected association is positive, as higher, volatile values of VIX suggest a greater (positive and rising) build-up of exchange market pressures.

The T10 yield serves as a relatively robust proxy for change in monetary policy shock from advanced economies, with the expected positive association with EMP. Given T10's high sensitivity, rising yields on US financial assets suggest a tightening of monetary policy and a resulting rise in the cost of global loanable funds as institutional investors and a shadow banking system proactively adjust their portfolios. A potential detrimental effect of a sudden policy move and capital markets' overreaction then exposes emerging

markets to unhedged positions and a loss of monetary policy independence (e.g. Rey 2014; Institute of International Finance (IIF) 2016).

Finally, and without much elaboration needed, fluctuations in global primary commodity prices (oil, sugar, ores and gold) are expected to have an inverse relationship with EMP. The higher the price, the higher are the export revenue windfalls, feeding domestic currency appreciation and a build-up of international reserves. As such, the overall EMP may be expected to decline.

In the baseline analysis, we carry out panel study tests for random- and fixed-effects, following some variables' scalability adjustments as shown below. Given potential endogeneity, the analysis focuses on the direction of associated changes. Analysis also includes robustness tests that try to limit large economy influence in the respective subgroup. An extended robust estimation of a panel VAR extends the baseline approach, with the results discussed in the next section.

Data

Our data on the country groups in Table 1 are monthly from January 2000 to September 2015. As documented in Table 2, data on exchange rates, international reserves (less gold), and an industrial production index come from the World Bank Global Economic Monitor (WB GEM). The IMF's International Financial Statistics provide the country-based domestic lending rate. Data on the bid-ask spread for each respective emerging market currency come from the currency information services provider OANDA. The US T10 and the VIX come from the Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis. Finally, information on commodity prices comes from UNCTAD's online statistical database commodity prices report.

Table 2 provides summary information on all variables, data sources and expected signs. Descriptive statistics for variables in each tested commodity group are provided in Tables A1, A2 and A3 of Appendix A. Unless otherwise noted in the results discussion, explanatory variables (except for VIX) are modified from the levels to either percentage change or to first difference.

IV. Empirical results analysis and some extensions to the base models

Baseline model results

We now turn to estimating the baseline model variants in Equation (1) for three commodity groups separately: sugar, ores and fuels. Estimation results for the sugar-exporting group of countries are reported in Table 3, for ores in Table 4, and for the fuels group in Table 5. For brevity, pretesting all variables with standard Im-

Pesaran-Shin and Fisher-type tests confirmed data stationarity, with the exception of T10, which was amended by taking the first difference.

Furthermore, initial panel estimations were conducted for standard ordinary least squares (OLS) and both random- and fixed-effects models. The Hausman test for partial and complete specification of Equation (1) points to the use of a random-effects model. This is intuitive for a mixed-sample model, and given the role of the commodity prices variable, it adds a random

Table 3. Sugar group EMP estimation: full sample period and pre/post price peak periods.

	Full sample period (Jan 2000-Sept 2015)				pre-price peak sample (Jan 2000 – Jan 2011)	post-price peak sample (Feb 2011- Sept 2015)
	(1)	(2)	(3)	(4)	(4)	(4)
VIX	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.002*** [0.0004]
dISUGAR	-0.048*** [0.019]	-0.03* [0.0194]	-0.03* [0.0194]	-0.029* [0.0194]	0.0043 [0.02309]	-0.0883** [0.0346]
dLR	0.006*** [0.002]	0.006*** [0.002]	0.006*** [0.002]	0.006*** [0.002]	0.009*** [0.003]	-0.004* [0.003]
dIOIL		-0.0791*** [0.0179]	-0.08471*** [0.0184]	-0.08462*** [0.0184]	-0.05916** [0.023]	-0.12357*** [0.0271]
dMPC		-0.0017 [0.0175]	-0.002 [0.018]	-0.003 [0.018]	-0.019 [0.022]	0.048** [0.024]
dT10			0.009 [0.007]	0.009* [1.27]	0.005 [0.008]	0.036*** [0.012]
dBIDASK				-0.0003 [0.0005]	-0.0004 [0.001]	-0.001 [0.001]
CONST	-0.0294*** [0.004]	-0.025*** [0.0041]	-0.0255*** [0.0041]	-0.0255*** [0.0041]	-0.0364*** [0.0054]	-0.0272*** [0.0065]
Obs.	940	940	940	940	660	280
Groups	5	5	5	5	5	5
St.Dev of residuals	0.04	0.04	0.04	0.04	0.05	0.03

***, **, * significance at 1, 5, and 10 per cent, respectively. Standard errors are in brackets. d in variable names refers to first difference; l are log-transformed variables.

Table 4. Ores group EMP estimation: full sample period and pre/post price peak periods.

	Full sample period (Jan 2000-Sept 2015)				pre-price peak sample (Jan 2000 – Jan 2011)	post-price peak sample (Feb 2011- Sept 2015)
	(1)	(2)	(3)	(4)	(4)	(4)
VIX	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0003]	0.001*** [0.0005]
dIORES	-0.2793*** [0.029]	-0.221*** [0.0314]	-0.238*** [0.0313]	-0.241*** [0.0313]	-0.228*** [0.03791]	-0.22194*** [0.05485]
dLR	0.009*** [0.002]	0.009*** [0.0019]	0.009*** [0.002]	0.009*** [0.002]	0.009*** [0.002]	0.004* [0.003]
dIOIL		-0.1005*** [0.0229]	-0.1253*** [0.0232]	-0.1231*** [0.0232]	-0.05797** [0.0285]	-0.2905*** [0.0373]
dMPC		-0.0408 [0.0435]	-0.032 [0.043]	-0.035 [0.043]	-0.056 [0.053]	0.021 [0.067]
dT10			0.039*** [0.008]	0.039*** [4.7]	0.03*** [0.01]	0.091*** [0.017]
dBIDASK				0.001* [0.0003]	0.001* [0]	0.003 [0.004]
CONST	-0.0239*** [0.0049]	-0.021*** [0.0049]	-0.023*** [0.0049]	-0.023*** [0.0049]	-0.0404*** [0.0063]	-0.0148* [0.0084]
Obs.	939	939	939	939	660	279
Groups	5	5	5	5	5	5
St.Dev of residuals	0.05	0.05	0.05	0.05	0.05	0.04

***, **, * significance at 1, 5, and 10 per cent, respectively. Standard errors are in brackets. d in variables names refers to first difference; l are log-transformed variables.

Table 5. Fuels group EMP estimation: full sample period and pre/post price peak periods.

	Full sample period (Jan 2000–Sept 2015)				pre-price peak sample (Jan 2000 – Jun 2008)	post-price peak sample (Jul 2008– Sept 2015)
	(1)	(2)	(3)	(4)	(4)	(4)
VIX	0.009 [0.0143]	0.009 [0.0144]	0.01 [0.0146]	0.01 [0.0147]	0.003*** [0.0008]	0.01 [0.0098]
dIOIL	0.2622 [1.37]	0.308 [1.4097]	0.221 [1.4667]	0.22 [1.4699]	−0.0144 [0.06548]	0.93746 [1.11886]
dLR	0.004 [0.038]	0.003 [0.038]	0.003 [0.038]	0.004 [0.038]	0.002* [0.001]	0.006 [0.014]
dIGOLD		−0.4531 [3.0634]	−0.3324 [3.1148]	−0.32754 [3.1219]	0.04012 [0.1368]	−0.30074 [0.3723]
dMPC		−0.0328 [0.5572]	−0.027 [0.558]	−0.018 [0.567]	−0.028 [0.022]	−0.047 [0.058]
dT10			0.119 [0.554]	0.118 [0.21]	0.058** [0.024]	0.027 [0.05]
dBIDASK				−0.0004 [0.0031]	−0.00039*** [0.0001]	0.01 [0.006]
CONST	−0.0762 [0.3261]	−0.079 [0.336]	−0.0872 [0.3707]	−0.0871 [0.3226]	−0.0871*** [0.0164]	0.0403 [0.2015]
Obs.	940	940	940	940	505	435
Groups	5	5	5	5	5	5
St.Dev of residuals	3.50	3.51	3.51	3.51	0.11	5.18

***, **, * significance at 1, 5, and 10 per cent, respectively. Standard errors are in brackets. d in variables names refers to first difference; l are log-transformed variables.

component to the balanced panel. Specifically, despite overlapping time periods, the commodity group designation effectively acts as a time-invariant constraint. Hence, results are reported for random-effects estimations only (robust standard errors' fixed-effects results, which largely support the reported findings, are suppressed).

Four model estimations are reported in each commodity group's results, with the last model (4) being most inclusive for our purposes (Table 3–5). We also report results for the pre- and post-commodity price peak time-period samples. The comparison offers insights as it allows the dissection of EMP effects between two distinct 'super-cycle' periods: the run-up to the commodity price peak and immediately afterwards.

Results in Tables 3 and 4 show some consistency in responses between the sugar and ores groups of exporters. This is irrespective of the unique, domestic factors and commodity group distinctions. We should note that both groups include countries with floating exchange rate regimes as per the IMF's report on exchange arrangements (International Monetary Fund (IMF) 2015). The key results here are consistent with the expected signs in Table 2 and the intuition in Aizenman and Hutchison (2012) that this paper regards as a baseline.

Starting with domestic factors, we note a positive relationship between the change in the domestic lending rate (dLR) and the EMP for both sugar and ores for the full sample and the pre-price peak sample period. Results indicate a possibly inverse effect in the sugar group during the post-price peak sample period. This is likely due to the stronger balance in external factors in the sugar-exporters' financial system and the weak financial deepening within the group.

The financial deepening argument is informative but very difficult to capture more precisely because of the lack of an adequate time series. Our intuition is confirmed in a recent study by Sahay et al. (2015) that shows the leading ores group economies as being more financially developed than those in the sugar group. Related is the difference in bid-ask spread (dBIDASK), which is significant and shows the expected signs in ores (unlike the sugar exporters group where BIDASK appears weaker). In a competitive foreign exchange market, a positive change in the bid-ask spread signals a heightened EMP index. A reserves loss is possible if the domestic central bank's active intervention relies on its trader power status (e.g., Khemraj and Pasha 2012). Such intervention might be feasible in an attempt to maintain the

currency peg, but that is if the shock feeding the banking system's bid-ask spread is considered temporary – a scenario that has played out in the post-GFC period.

The change in the domestic industrial index dMPC is negatively correlated with the EMP for both sugar and ores in the full and pre-peak price periods, but the coefficient reverses its sign for both in the post-peak price period. The latter is yet another transformation of the factors shaping the EMP in the post-GFC period. A possible interpretation, again, points to the role more speculative financial factors play, as the CDCs are perceived as risky investments. Hence, a perception of a negative shock in the post-GFC period exacerbates exchange market pressures despite the real economy's industrial improvement.

The results also report a direct correlation between the emerging markets' EMP and the global volatility (VIX) and liquidity gauges (dT10). A possible interpretation could be that as volatility and spreads on sovereign yields rise, countries in both commodity groups are expected to face rising borrowing costs and abrupt pressures on domestic currency. The response would then vary, based on an individual country's exchange rate policy and use of capital controls – a point addressed in Fernández et al. (2015).

Perhaps most curious are the results related to the changes in the log difference of the underlying commodity's price (dISUGAR or dLORES) and the price of oil (dIOIL). The latter serves as a gauge for global commodity demand and plays a dominant role in both groups. With the exception of the price of sugar in the pre-price peak period, the results are as expected, with a rise in the core commodity price leading to lower EMP index, which, as discussed earlier, feeds into currency appreciation and reserves accumulation.

In Appendix B, Tables B1 and B2 provide an alternative estimation, as a robustness check, for the sugar and ores groups (suggested by one of the reviewers of an earlier draft of this paper). Eliminating observations for Brazil from the sugar panel and Chile from the ores panel helps to control for a large economy effect. The results mostly agree with the original findings, with a re-

emphasizing of this paper's initial expectations about the relation between the explanatory factors and the EMP with two caveats.

First, Table B1 reports the lending rate in the sugar group as being positive and significant in the full sample model (4). However, dLR remains consistently negative and significant in the post-price peak sample. The reversal may be an indication of the financial condition's influence over the real economy in the case of the sample's remaining countries. This is reconfirmed in the similar reversal in the dMPC coefficient sign and relatively larger coefficient for the oil price change.

The second observation is the relative heterogeneity of the sugar group and the somewhat relatively homogenous ores group. In the first, with 16.3% of the global exports share, Brazil dominates the smaller sugar exporters. On the other hand, in the ores group, Chile's 5.5% global ores exports share is much closer to other economies in the group, which explains the robustness of the results for the ores sample in Table B2.

The above brings us to the fuel group results (Table 5) that are distinct from the other two. According to Table 5 findings (verified separately in other panel extensions), the signs seem to follow the expected direction in most cases, yet the majority of the coefficients are not statistically significant. What matters is not just whether a commodity is designated as being renewable (usually agricultural products but also new types of energy, e.g., wind and hydroelectric power) or non-renewable (typically metals, coal, natural gas, crude oil, etc.), but of critical importance is the country's chosen exchange rate policy, vis-à-vis its macroeconomic fundamentals, and the type of commodity dependence.

A type of announced foreign exchange regime may play a significant role in explaining these results. As indicated earlier in Table 1, the countries comprising this group have actively maintained a type of currency peg (with the recent exception of Russia within the sample period). This may in fact explain EMP's different dynamic, as policy proactively intervenes to offset any domestic or external pressures. Testing this intuition, we report in Table 6, percentage-change results for month-on-month international reserves

Table 6. Fuels group RESERVES change estimation.

	Full sample period (Jan 2000–Sept 2015)	pre-price peak sample (Jan 2000 – Jun 2008)	post-price peak sample (Jul 2008–Sept 2015)
	(4)	(4)	(4)
VIX	−0.001*** [0.0004]	−0.002*** [0.0007]	−0.001 [0.0004]
dIOIL	0.045 [0.0376]	0.019 [0.0598]	0.021 [0.046]
dMPC	0.028* [0.015]	0.041** [0.021]	0.003 [0.022]
dBIDASK	0.0004*** [0.0001]	0.00037*** [0.0001]	0.00146 [0.0025]
dLR	−0.0005 [0.001]	0.0001 [0.001]	0.001 [0.002]
dT10	−0.0362** [0.0145]	−0.055*** [0.022]	−0.01 [0.018]
CONST	0.0425***	0.071***	0.0123
Obs.	940	505	435
Groups	5	5	5
St.Dev of residuals	0.09	0.10	0.08

***, **, * significance at 1, 5, and 10 per cent, respectively. Standard errors are in brackets. d in variables names refers to first difference; l are log-transformed variables.

in response to the domestic and external factors in the full sample period and in the pre-price peak and post-price peak samples.

As expected, the international reserves are inversely related to the main external risk factors (VIX and T10), as countries try to shore up more foreign currency when capital flows intensify. As Semmler and Gevorkyan (2011) note, reserves can then be used to either (a) support domestic private credit flows, or more likely, (b) maintain the exchange-rate peg. However, the second option suggests that the external shock is perceived as transitory. Hence, there is also a correlated rise in reserves following positive changes in the price of oil and industrial production. The results for the two financial domestic factors, dBIDASK and dLR, are significant in the former case and lacking significance in the latter case but are also initially counter to the implied expectations for a change in international reserves.

Given possible endogeneity in the model, one may suggest that Table 6 also reports on correlation across changes of dBIDASK, dLR and reserves. In this case, a rise in the uncertainty about exchange rates or the first signs of foreign exchange pressures, inferred in positive dBIDASK and dLR changes, contribute to the same precautionary behaviour of shoring up international reserves as the immediate response. This derivation is further confirmed in the panel VAR analysis below.

All in all, in the fuels group the monetary authority is capable of withstanding a mix of domestic and external factor pressures, as long as the international reserves cushion allows for it. The question remains: what, if anything, does the fuels group exporters stand to gain from a fixed or pegged exchange rate (and what do they stand to lose)? This is yet to be seen as the global trade and financial patterns transform with policy tightening.

Critically, the accumulation and replenishment of the international reserves cushion in the CDCs is a function of strong global demand for the underlying commodity (in which China is a key driver, as observed by a reviewer of an earlier draft) and a rising primary commodity price. To the extent that these economies represent a perceived opportunity of higher return on investment (e.g. with average GDP per capita growth rates), there may be a net foreign currency inflow (as in Institute of International Finance (IIF) 2016). That eases exchange market pressures but is not reflected in the domestic and external factors, although VIX, MPC, BIDASK and T10 lend some support to such a view. As such, the dynamic in the fuels groups of countries appears to be more consistent with the surges in capital flows analysis in the recent work by Ghosh et al. (2012). At the same time, structural characteristics, such as the lacking of industrial diversification and the dominance of energy exports, help explain the preference for a clearly observed exchange rate target.

A robustness check and a possible extension

In addition to the robustness estimation presented in Tables B1 and B2 of the Appendix B, the core results are further confirmed by estimating a panel VAR model. For compactness, only orthogonalised impulse response (ORFIs) graphs are included in this discussion. The tests were carried by applying the pvar routine in Stata.

Abrigo and Love (2015) offer a detailed theoretical and practical introduction to the pVAR procedures. The key feature of this modelling technique is capturing one period lagged effect in a panel study that is applicable in the emerging markets case of this paper. As such, the tests were based on a generalized method of moments

(GMM) estimation. The impulse response functions were derived upon establishing the estimated panel VAR's stability. The confidence intervals were computed using a Monte Carlo simulation based on the estimated model. The ORFIs allow for causality, although much caution should be exercised due to great data fragility. Other explanatory variables play a role, as well.

The results for the three commodity groups appear in Appendix C in Figures C1 through C6. Note that for each commodity group there are two sets of graphs: showing domestic and external factors. Each set includes three figures (a, b and c) with the exchange market pressure (EMP), the main variable of interest, the percentage change in international reserves [the second component in the EMP (RESchng)], and the percentage change in the exchange rate [the first component in the EMP (FXchng)]. In other words, following equation (2), the RESchng is defined as $\frac{R_{it}-R_{it-1}}{R_{it-1}}$ and FXchng as $\frac{e_{it}-e_{it-1}}{e_{it-1}}$. It is informative to review such EMP decomposition interacting with the key factors of the main model in each commodity group (and exchange regime).

In the sugar group, Figure C1(a) shows a positive one-period lagged EMP response to the change in industrial production (dMPC) shock. Initially uneven, the result is largely consistent with the earlier findings. This seems to be further reconfirmed in panel (b), which reports positive change in reserves as a response to the dMPC. At the same time, it is evident from panel (c) that there is a downward pressure (depreciation) on the currency as the exchange rate change response is positive, with a positive dMPC shock.

For the smaller economies in the sugar group external factors have more force in determining the EMP direction, which also has been confirmed in the earlier estimates (Table B1). In Figure C2 of Appendix C, the reserves respond strongly to a positive shock to the price of sugar, driving overall sugar EMP down. RESchng responds in similar fashion to the VIX and dT10 shocks, while the response of the FXchng is the opposite. Both outcomes suggest a high degree of the sugar group's perceptiveness to the global economy trends, consistent with the exporter economies' relative size. Finally, the shock to dBIDASK results in a positive

change in reserves, which may be explained as a central bank's proactive response to mounting foreign exchange pressures as evidenced in the overall EMP decline.

There is less initial ambiguity in the ores' EMP negative response to a positive shock to the commodity price (Appendix C Figures C3 and C4). Similarly, ores' EMP responds as expected, releasing the pressure, which results in improving the industrial production index. Reserves adjust positively to the dLORES shock and much more strongly compared to the exchange rate changes in both domestic and external factor panels. The external ORFIs do not include the price of oil. This is done to maintain focus on commodity exporter-specific factors, despite the dominating role of the price of crude oil remains. The ores group results are also consistent with the earlier panel findings, explained here as a precautionary reserves build-up that significantly exceeds responses in the exchange rate. From the external perspective, shocks to the VIX and dT10 seem to lead to an expected rising EMP tension with one lag with implied non-linearity of capital markets.

The fuels group EMP results (Appendix C Figures C5 and C6) reconfirm those in Tables 5 and 6 earlier. There is a limited response from EMP in the domestic and external panels. Decomposing, ORFIs clearly show a positive response in reserves to the positive shock in the price of oil and subsequent adjustment. This is explained by the implied pegged exchange rate policy, with no change to the FXchange response. This finding further suggests that RESchng is the leading variable in the EMP definition for the fuel country group, operating with a currency peg regime.

It is instructive to keep the international reserves factor in mind for all three commodity groups and for all currency regimes. With varying intensity (dependent upon macroeconomic structure), reserves tend to rise in response to the positive shocks of the main commodity price across the three groups. In sugar and ores, RESchng responds positively to widening bid-ask spreads in the foreign exchange markets. Such outcomes may be due to heightened perceptions of risk and precautionary moves by central banks to shore up foreign currency in anticipation of a sudden reversal of capital flows.

V. Conclusion

For many primary commodity-exporting emerging markets, macroeconomic and social development is directly linked to the volatility in the price of their primary commodity export. This paper analysed the short-term sensitivity of foreign exchange regimes in commodity-dependent economies. The analysis focused on the exchange market pressure for three groups of commodity-exporting emerging markets for pre- and post-price peak periods. For added discussion, the paper developed a panel VAR model that captured lagged effects decomposing exchange rate and international reserves changes across the sample groups. The main findings have strong macroeconomic policy implications for the CDCs in balancing between maintaining a competitive exchange rate and attempting structural development.

The paper uncovers a substantial degree of heterogeneity across commodity groups and exchange regimes with domestic and external factors playing an interchangeable role affecting the EMP. Looking at the post-peak price results, it is evident that much of the post-crisis resilience, in particular across fuel-exporting countries, has been due to a reliance on international reserves accumulation with a proactive central bank intervention. At the same time, the domestic lending rate and bid-ask spread on foreign currency pointed to an induced currency depreciation. The conclusions are confirmed also for short-term external factors (VIX and T10).

The results also suggest that in countries with floating exchange regimes (sugar and ores groups), price volatility of the primary commodity, which impacts the terms of trade, may also have implications for foreign currency denominated debt sustainability as demand for foreign currency rises, as shown in this paper in higher EMP (e.g. Basu and Gavin 2011). In the fixed exchange regimes, the external and domestic pressures lead to a loss in reserves.

As mentioned, individual country outcomes will vary. For instance, it is unclear to what extent reliance on reserves is manageable, as monetary authorities run out of effective policy options. As FDI follows competitive markets, signs of export

revenue strain in the background of declining commodity prices may contribute to higher borrowing costs in international capital markets and further volatility in EMP. The confluence of these factors feeds into the emerging markets CDC's rising social costs and post-GFC macroeconomic uncertainty.

The analysis carried out in this paper offers a multifaceted approach. Contributing to the literature on the exchange market pressure, this paper also attempts to pierce beyond the respective commodity price pre-peak and post-peak patterns of domestic and external driving forces. The role of currency regimes, aligned by groups and studied in the paper, suggests country-specific heterogeneity, which may be explored in a possible extension. Overall, results are informative for a development policy's macroeconomic measures design and for the short-term proactive foreign exchange market adjustments by the central banks in a diverse mix of countries.

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No potential conflict of interest was reported by the author.

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Appendixes

Appendix A

Table A1. Country code-country name key.

Country code	Country name	Country code	Country name	Country code	Country name
DZA	Algeria	CIV	Côte d'Ivoire	RUS	Russia
AGO	Angola	GHA	Ghana	ZAF	South Africa
ARG	Argentina	GTM	Guatemala	THA	Thailand
BRA	Brazil	IND	India	UKR	Ukraine
CHL	Chile	NGA	Nigeria	VEN	Venezuela
COL	Colombia	PER	Peru		

Table A2. Summary Descriptive Statistics for SUGAR group.

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Group specific: domestic</i>					
EMP	945	(0.01)	0.05	(0.43)	0.28
MPC	945	0.00	0.05	(0.35)	0.37
BIDASK	945	0.26	2.71	(0.96)	62.39
LR	945	18.89	15.66	5.50	74.20
<i>External</i>					
OIL	945	64.79	31.15	18.53	132.50
SUGAR	945	13.78	6.18	5.14	29.61
VIX	945	20.78	8.24	10.42	59.89
T10	945	3.74	1.20	1.53	6.66

Variables in levels format.

Table A3. Summary Descriptive Statistics for ORES group.

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Group specific: domestic</i>					
EMP	945	(0.01)	0.06	(0.18)	0.30
MPC	945	0.00	0.02	(0.17)	0.22
BIDASK	945	(2.10)	7.26	(50.64)	(0.00)
LR	945	20.52	15.49	3.54	74.20
<i>External</i>					
OIL	945	64.79	31.15	18.53	132.50
ORES	945	3,463.43	1,574.51	1,051.66	6,633.20
VIX	945	20.78	8.24	10.42	59.89
T10	945	3.74	1.20	1.53	6.66

Note: Variables in levels format.

Table A4. Summary Descriptive Statistics for FUELS group.

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Group specific: domestic</i>					
EMP	945	0.11	3.49	(1.47)	107.23
MPC	945	0.01	0.13	(0.46)	0.84
BIDASK	945	1.05	26.53	(1.00)	813.06
LR	945	21.00	21.15	7.90	125.97
<i>External</i>					
OIL	945	64.79	31.15	18.53	132.50
GOLD	945	841.23	479.52	260.48	1,771.85
VIX	945	20.78	8.24	10.42	59.89
T10	945	3.74	1.20	1.53	6.66

Variables in levels format.

Appendix B

Table B1. Sugar group EMP estimation: full sample period and pre/post price peak periods, excluding Brazil.

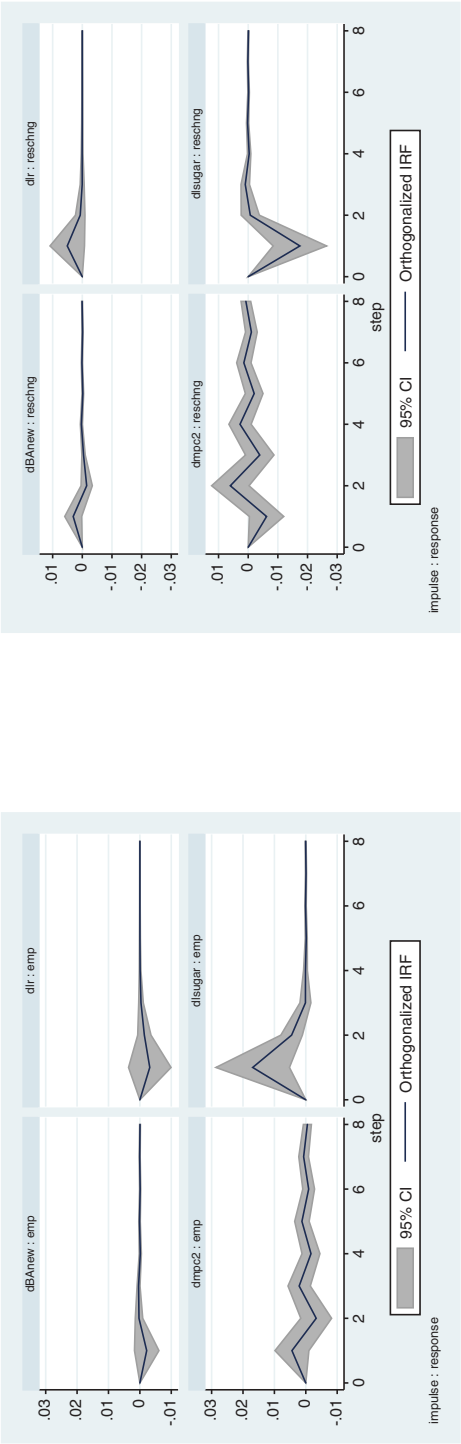
	Full sample period (Jan 2000-Sept 2015)				pre-price peak sample (Jan 2000 – Jan 2011)	post-price peak sample (Feb 2011- Sept 2015)
	(1)	(2)	(3)	(4)	(4)	(4)
VIX	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.002*** [0.0004]
dISUGAR	–0.0564*** [0.018]	–0.044** [0.0187]	–0.044** [0.0187]	–0.043* [0.0187]	–0.0225 [0.0219]	–0.04749 [0.03693]
dLR	–0.006* [0.004]	–0.006* [0.0043]	–0.006* [0.004]	–0.006* [0.004]	–0.005 [0.005]	–0.018** [0.008]
dIOIL		–0.0541*** [0.0172]	–0.05684*** [0.0176]	–0.05666*** [0.0176]	–0.03475* [0.0218]	–0.09297*** [0.0284]
dMPC		0.0052 [0.0152]	0.005 [0.015]	0.004 [0.015]	–0.009 [0.019]	0.047** [0.023]
dT10			0.005 [0.007]	0.005 [0.69]	0.001 [0.008]	0.032*** [0.013]
dBIDASK				–0.0005 [0.0005]	–0.0006 [0.001]	–0.001 [0.001]
CONST	–0.0231*** [0.0038]	–0.02*** [0.004]	–0.0203*** [0.004]	–0.0202*** [0.004]	–0.0285*** [0.0051]	–0.0292*** [0.0069]
Obs.	752	752	752	752	528	224
Groups	4	4	4	4	4	4
St.Dev of residuals	0.04	0.04	0.04	0.04	0.04	0.03

***, **, * significance at 1, 5, and 10 per cent, respectively. Standard errors are in brackets. d in variable names refers to first difference; l are log-transformed variables.

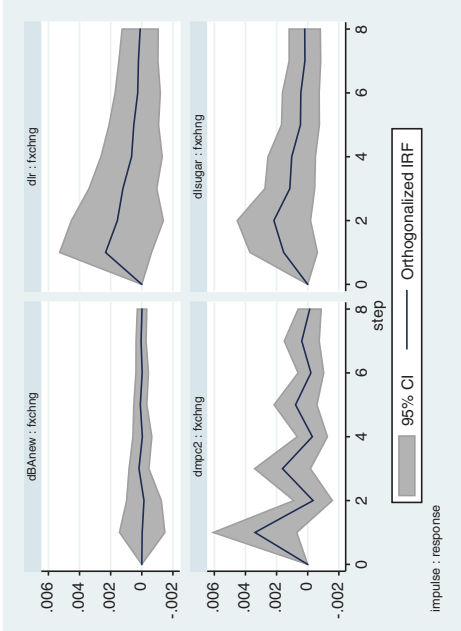
Table B2. Ores group EMP estimation: full sample period and pre/post price peak periods, excluding Chile.

	Full sample period (Jan 2000-Sept 2015)				pre-price peak sample (Jan 2000 – Jan 2011)	post-price peak sample (Feb 2011- Sept 2015)
	(1)	(2)	(3)	(4)	(4)	(4)
VIX	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.001*** [0.0002]	0.002*** [0.0003]	0.001*** [0.0005]
dIORES	–0.2835*** [0.032]	–0.221*** [0.0352]	–0.239*** [0.035]	–0.242*** [0.0351]	–0.2142*** [0.0424]	–0.24603*** [0.0594]
dLR	0.011*** [0.002]	0.01*** [0.0021]	0.01*** [0.002]	0.01*** [0.002]	0.01*** [0.003]	0.006** [0.003]
dIOIL		–0.1088*** [0.0257]	–0.13463*** [0.0261]	–0.1324*** [0.0261]	–0.0657** [0.0319]	–0.29786*** [0.041]
dMPC		–0.036 [0.0504]	–0.023 [0.05]	–0.028 [0.05]	–0.053 [0.063]	0.02 [0.07]
dT10			0.041*** [0.009]	0.041*** [4.38]	0.033*** [0.011]	0.083*** [0.018]
dBIDASK				0.001* [0.0004]	0.001* [0]	0.004 [0.004]
CONST	–0.0282*** [0.0055]	–0.025*** [0.0055]	–0.0273*** [0.0054]	–0.0272*** [0.0054]	–0.0484*** [0.0071]	–0.017* [0.0091]
Obs.	752	752	752	752	528	224
Groups	4	4	4	4	4	4
St.Dev of residuals	0.05	0.05	0.05	0.05	0.05	0.04

***, **, * significance at 1, 5, and 10 per cent, respectively. Standard errors are in brackets. d in variable names refers to first difference; l are log-transformed variables.



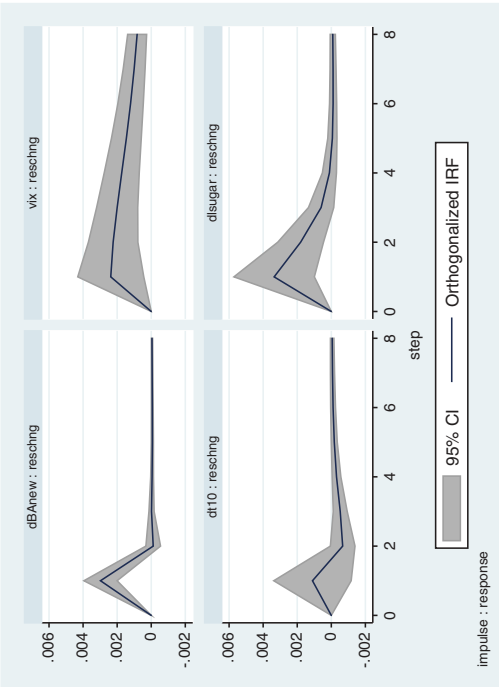
(a) EMP response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).



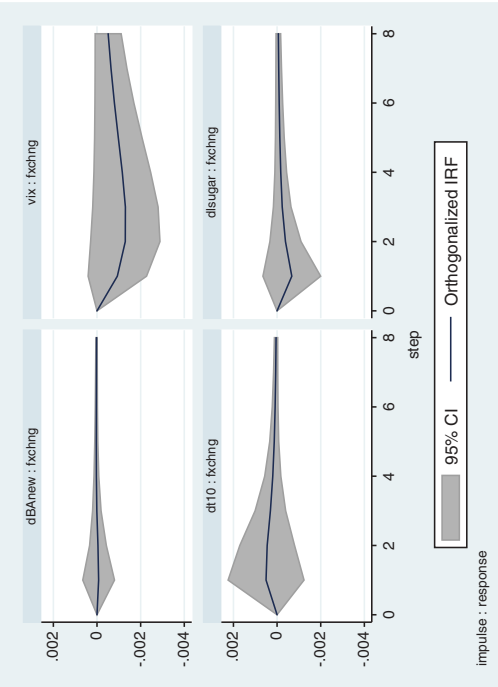
(b) International reserves response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).

(c) Exchange rate response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).

Figure C1. SUGAR: domestic factors panels.



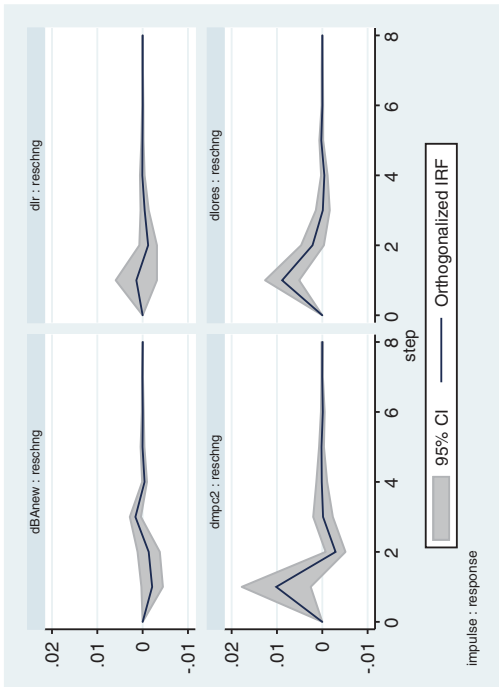
(a) EMP response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).



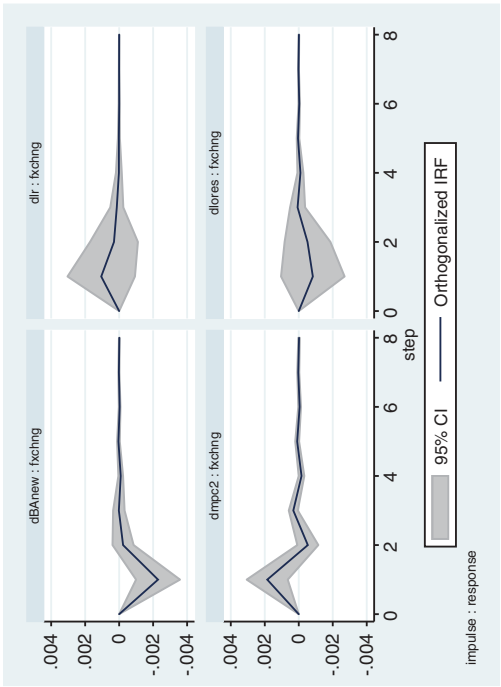
(b) International reserves response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).

(c) Exchange rate response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).

Figure C2. SUGAR: external factors panels.

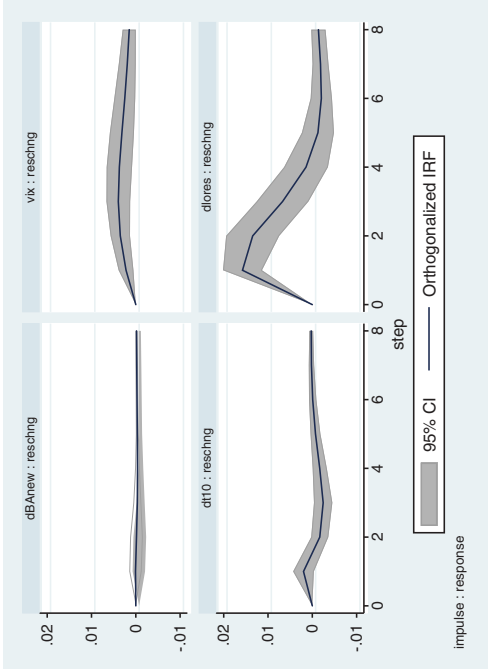


(a) EMP response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).

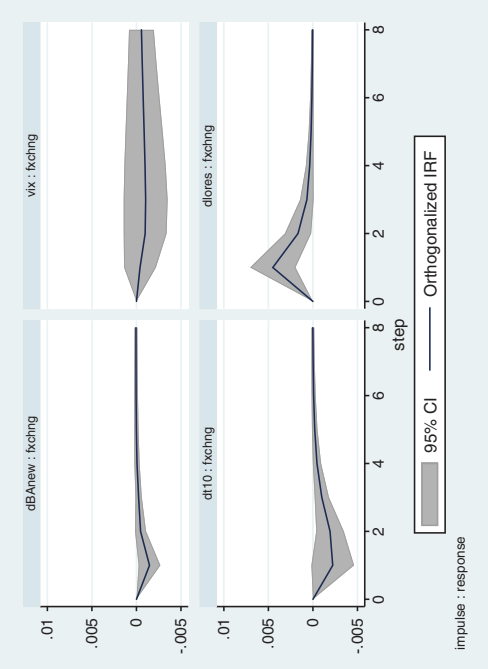


(b) International reserves response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).

Figure C3. ORES: domestic factors panels.



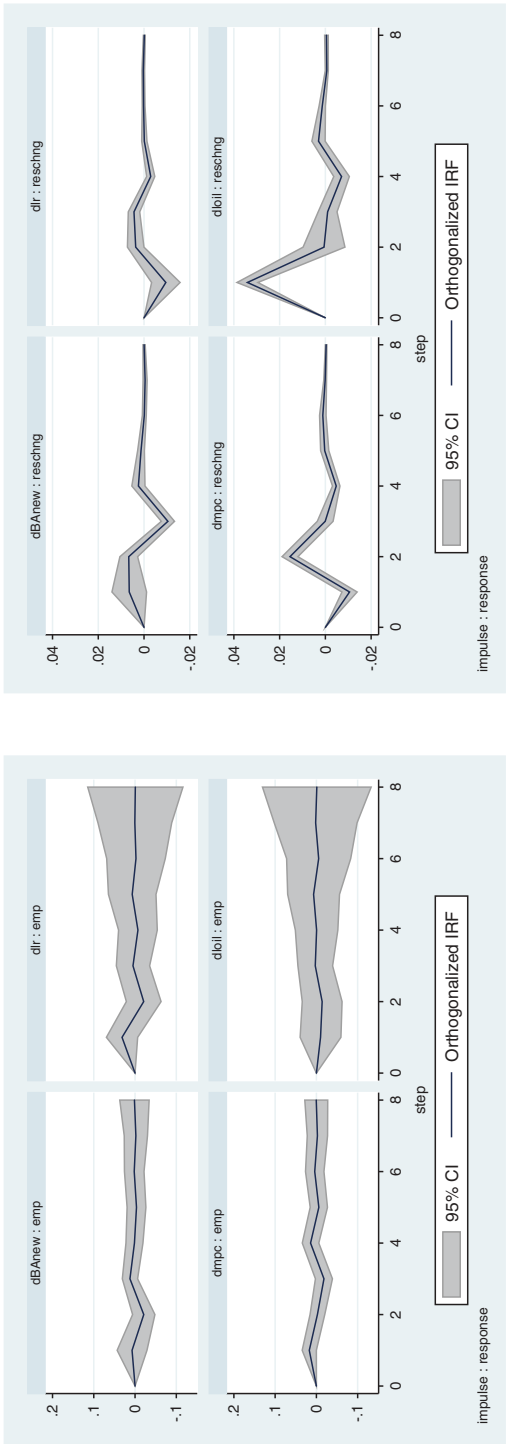
(a) EMP response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).



(b) International reserves response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).

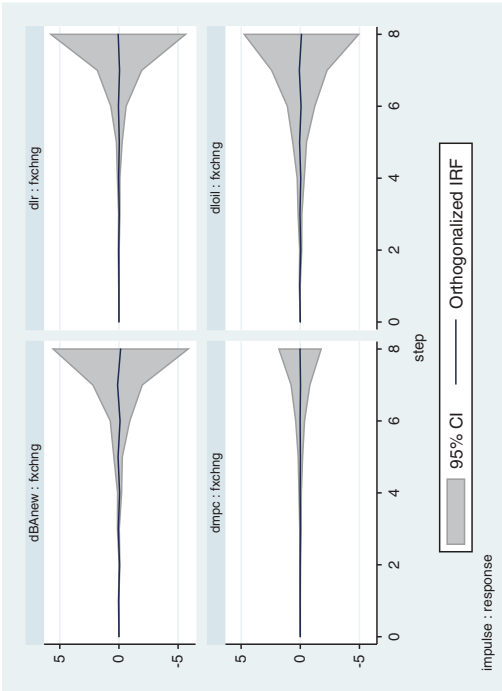
(c) Exchange rate response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).

Figure C4. ORES: external factors panels.



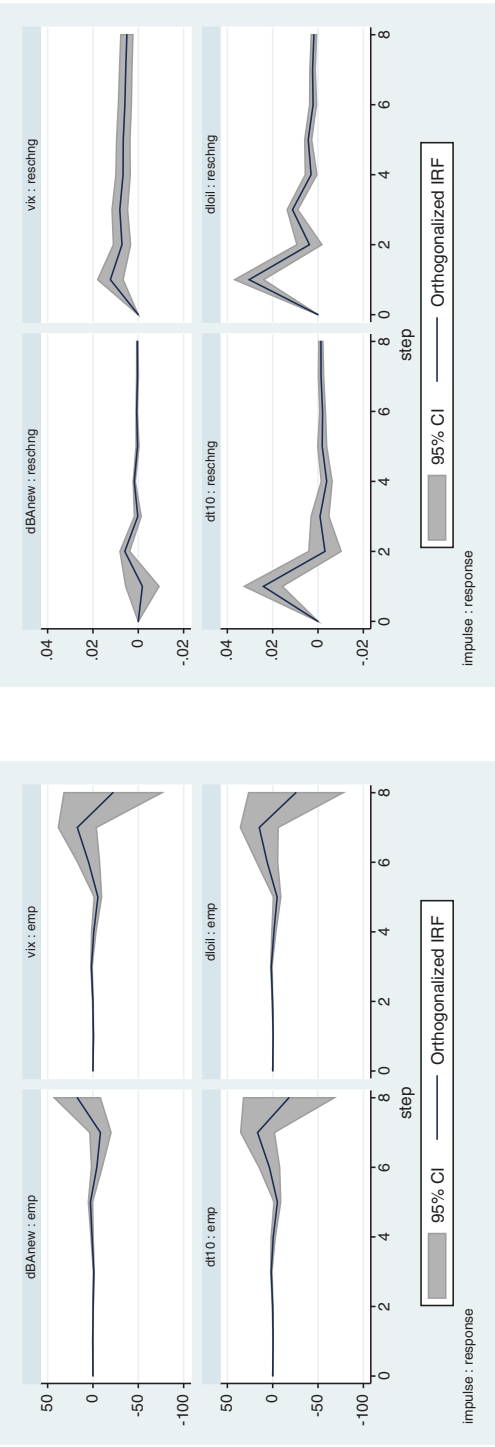
(a) EMP response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).

(b) International reserves response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).



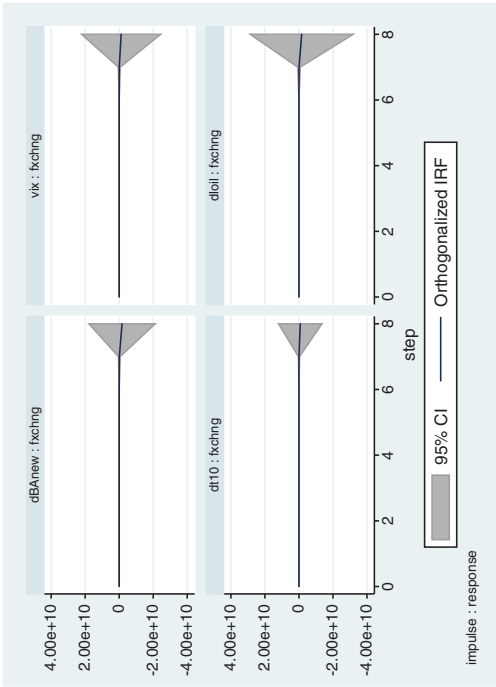
(c) Exchange rate response to shock from bid-ask spread, lending rate, commodity price, and industrial production (clockwise).

Figure C5. FUELS: domestic factors panels.



(a) EMP response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).

(b) International reserves response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).



(a) Exchange rate response to shock from bid-ask spread, VIX, commodity price, and Treasury-10 (clockwise).

Figure C6. FUELS: external factors panels.